Tutorial No : 9

Tutorial Title: Represent graphs and generate minimum spanning trees using Prim's algorithm.

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Program:

```
#include <stdio.h>

#define MAX 20
#define INF 9999

int main()
{
   int cost[MAX][MAX], visited[MAX] = {0};
   int n, i, j, edges = 0;
   int min, a, b, total = 0;

   printf("Enter number of vertices: ");
   scanf("%d", &n);

   printf("Enter adjacency matrix:\n");
   for (i = 0; i < n; i++)
   {
      for (j = 0; j < n; j++)
      {
         scanf("%d", &cost[i][j]);

         if (cost[i][j] == 0)
            cost[i][j] = INF;
      }
   }

   visited[0] = 1;

   printf("Edges in Minimum Spanning Tree:\n");

   while (edges < n - 1)
   {
```

```
        min = INF;

        for (i = 0; i < n; i++)
        {
            if (visited[i])
            {
                for (j = 0; j < n; j++)
                {
                    if (!visited[j] && cost[i][j] < min)
                    {
                        min = cost[i][j];
                        a = i;
                        b = j;
                    }
                }
            }
        }

        if (min == INF)
        {
            printf("Graph is not connected.\n");
            return 0;
        }

        printf("%d - %d : %d\n", a + 1, b + 1, min);
        total += min;
        visited[b] = 1;
        edges++;
    }

    printf("Total cost of MST = %d\n", total);

    return 0;
}
```

Output;

Output

```
Enter number of vertices: 4
Enter adjacency matrix:
0 2 0 6
2 0 3 8
0 3 0 7
6 8 7 0
Edges in Minimum Spanning Tree:
1 - 2 : 2
2 - 3 : 3
1 - 4 : 6
Total cost of MST = 11
```